

Fig. 2.7 In a "processed method" the inputs are first transformed before repeated application of a given method; after post-processing, the output may have higher order of accuracy than would be obtained using a straightforward application of the scheme

The use of conjugacy as a method to develop enhanced numerical methods was introduced in [397] under the name *symplectic correctors*.

2.5 Other Classes of Methods

2.5.1 Runge-Kutta Methods

The family of Runge-Kutta methods for solving $\dot{z} = f(z)$ is defined by

$$Z = z + h \sum_{i=1}^{s} b_i F_i,$$

where the vectors F_i, $i = 1, \ldots, s$, are computed by solving the system

$$F_i = f(z + h \sum_{j=1}^{s} a_{ij} F_j), \qquad i = 1, \ldots, s.$$

The parameters s, $b_1, b_2, \ldots, b_s$, and the $s \times s$ matrix of coefficients $A = (a_{ij})$ describe the method. In some cases the determination of the vectors F_i requires the solution of a nonlinear system, i.e. the method is *implicit*; in other cases the method can be explicit. An example of a popular 4th order explicit method is the choice of matrix A with coefficients $a_{ij} \equiv 0$ except $a_{21} = 1/2$, $a_{32} = 1/2$ and $a_{43} = 1$, and $b_1 = 1/6$, $b_2 = 1/3$, $b_3 = 1/3$, $b_4 = 1/6$. This method is not symplectic, and it is in fact impossible to find symplectic explicit methods within the Runge-Kutta family.

Let us emphasize that, while a typical RK method is not symplectic, **some** implicit Runge-Kutta methods **are** symplectic. The precise condition that must be